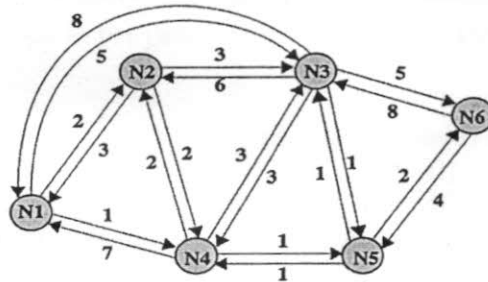


14. Traditional Telecom network implements _____ switching technique
 A) Circuit-switched B) Datagram
 C) Virtual-circuit D) None of the above
15. A crossbar switch with x inputs and y outputs requires _____ crosspoints
 A) xy B) x^2
 C) x/y D) y^2
16. What determines the packet delivered to a destination via the best path?
 A) Routing Algorithm B) IP Address
 C) Source Address D) MAC Address
17. The packet sent by a node to the source to inform it of congestion is called _____
 A) Explicit B) Discard
 C) Choke D) Backpressure
18. In the traffic model, the data rate changes in a very short time.
 A) Constant bit rate B) Variable bit rate
 C) Bursty D) None of these
19. In wireless ad-hoc network _____
 A) access point is not required B) nodes are not required
 C) access point is must D) all nodes are access points
20. The maximum size of payload field in ethernet frame is _____
 A) 1000 bytes B) 1300 bytes
 C) 1200 bytes D) 1500 bytes

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Define Data Communication. With neat diagram, explain simplified communication model of computer network.	5	L *2	1	1
	b) Explain four categories of noises. In a communication channel, the power of a signal is 10 mW and the power of the noise is 1μW; what are the values of SNR and SNR _{dB} ?	6	L 2	2	1,2
	c) What is the purpose of FSK? Explain how BFSK is implemented, with suitable diagram.	5	L 2	2	1,2
2.	a) Illustrate Encapsulation of data in a network model. Explain the concept of PDU in TCP/IP.	5	L 2	1	1
	b) Explain the digital signal encoding method adopted in Manchester encoding technique with suitable example. Mention its advantages and disadvantages.	5	L 2	2	1
	c) With block diagram, explain error correction process. A code scheme has a Hamming distance $d_{min} = 6$. i) What is its error detection capability? ii) What is its error correction capability?	6	L 2	2	1,2
3.	a) Draw the TCP/IP Protocol Suite by specifying protocols of each layer. Summarize the functionalities of every layer of the TCP/IP model.	8	L 2	1	1
	b) Compute the following by applying CRC with divisor 10011. i) Find the codeword corresponding to sequence of data 1101011011. ii) If the codeword has error in third bit (from the left), what does receiver obtain when it does its error checking?	8	L 3	2	1,2
Unit – II					
4.	a) Show the pictorial representation of sender window; briefly explain the concept of sliding window in flow control.	5	L 2	3	1,2
	b) Briefly explain multiplexing and de-multiplexing process in FDM, with suitable diagram.	6	L 2	3	1,2
	c) Compare circuit switching and packet switching networks.	5	L 2	4	1,2

5. a) Explain how two stations exchange data in CDMA protocol. Consider a general approach. 5 L 2 3 1,2
 b) Summarize the difference between Go-Back-N and Selective Repeat ARQ protocol. 6 L 2 3 1,2
 c) Explain the technique adopted by FHSS to protect the data from an intercepting eavesdropper. 5 L 2 3 1,2
6. a) Explain Synchronous TDM. Consider four 1-kbps connections are multiplexed together. A unit is 1 bit. Compute (i) the duration of 1 bit before multiplexing, (ii) the transmission rate of the link, (iii) the duration of a time slot, and (iv) the duration of a frame. 8 L 3 3 1,2
 b) Write pseudocode for Dijkstra's algorithm to find least cost path. Apply the same for the given network below, by considering 'N1' as source node. Provide detailed step by step evaluation of the best route.



8 L 3 4 1,2

Unit – III

7. a) Define traffic shaping. Explain leaky bucket algorithm for traffic shaping/traffic policing, with a neat diagram. 8 L 2 5 1,2
 b) With a flowchart, explain IEEE 802.11 Medium Access Control Logic. 8 L 2 5 1,2
8. a) Explain three persistent methods implemented in CSMA protocol, with suitable diagram. 8 L 2 5 1,2
 b) Show the IEEE802.11 MAC Frame format. Explain each fields in brief. 8 L 2 5 1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
